

Dataset Overview

This project analyzes the Titanic passenger dataset, which contains records for 891 passengers aboard the RMS Titanic, including demographic information (age, sex), ticket details (class, fare, cabin), family relationships aboard (siblings/spouses, parents/children), and the outcome of interest: whether each passenger survived the disaster. The purpose of this analysis is to act as a Junior Data Analyst investigating patterns in survival, cleaning and preparing the data for further use, and surfacing business-relevant insights that could inform future predictive modeling work.

Data Cleaning

The raw dataset contained several data quality issues that needed to be addressed before analysis. The Age column had 177 missing values, which were filled using the median age rather than the mean, since median is more resistant to the influence of outliers. The Embarked column had 2 missing values, filled using the mode (the most frequently occurring embarkation port). The Cabin column was dropped entirely, as the vast majority of its values were missing, making imputation unreliable. No duplicate rows were found in the dataset. Finally, several columns (Survived, Pclass, Sex, Embarked) were converted to categorical data types, since they represent a fixed, limited set of categories rather than continuous numeric values, improving both memory efficiency and analytical clarity.

Feature Engineering

Several new features were engineered to enable deeper analysis. AgeGroup categorizes passengers into Child, Teen, Adult, and Senior buckets, allowing survival patterns to be examined across life stages rather than raw age values. FamilySize combines the SibSp (siblings/spouses) and Parch (parents/children) columns plus one (for the passenger themselves), representing the total family unit size traveling together. TravelAlone flags whether a passenger had a FamilySize of exactly one, isolating solo travelers. FareCategory buckets the continuous Fare column into Low, Medium, and High groups for easier comparison. Finally, Title was extracted from each passenger's Name field using pattern matching, isolating honorifics such as Mr, Mrs, Miss, and rarer titles like Countess or Reverend, which carry social and demographic signal beyond raw name text.

Key Findings

The analysis revealed several consistent, strong patterns. Gender was the single strongest predictor of survival: female passengers survived at a rate of 74.2%, compared to just 18.9% for male passengers. Passenger class showed a similarly strong effect, with first-class passengers surviving at 63.0% compared to only 24.2% for third-class passengers. Fare, closely tied to class, showed the same pattern: passengers in the "High" fare category survived at 77.0%, versus just 27.8% for the "Low" fare category. Family size also mattered, with passengers traveling with a small family (2-4 members) surviving more often than those traveling entirely alone or in very large family groups. Finally, embarkation port showed a notable difference, with Cherbourg passengers surviving at 55.4%, compared to 33.9% for Southampton passengers, likely explained by a higher concentration of first-class passengers boarding at Cherbourg.

Business Insights

For a shipping or passenger transport company today, these findings suggest several practical takeaways beyond the historical context. Emergency preparedness and evacuation procedures should ensure equitable access across all ticket classes, not favor higher-paying passengers, both for ethical reasons and modern regulatory compliance. Understanding that family groups of moderate size navigated emergencies more successfully than solo travelers or very large groups could inform crew training on group evacuation assistance. Additionally, this analysis demonstrates the value of maintaining clean, well-structured passenger data, since the ability to quickly segment and analyze passengers by class, fare, and family structure proved essential to uncovering these patterns.

Future Machine Learning Opportunities

This cleaned and feature-engineered dataset is well-suited for a future predictive modeling project, specifically building a classification model to predict passenger survival based on pre-disaster characteristics. Features such as Pclass, Sex, Age, Fare, and FamilySize appear to carry genuine predictive signal without leaking post-outcome information, making them reasonable candidates for a first modeling attempt. Care should be taken with the Cabin feature, given its high missingness, and its "missingness" pattern itself should be investigated as a potential indirect signal before inclusion. A logistic regression or decision tree model would likely serve as a strong, interpretable baseline for this prediction task before considering more complex approaches.